

TopoRec-Family: Adaptive Reasoning Topology for Candidate-Aware LLM Recommendation

A minimal-input reranking protocol: topology changes the organization of evidence, not the available information.

1 Inputs

User History H_u
recent interacted items
item titles; optional ratings/time

Base Retriever R
SASRec / CRM / LightGCN
full catalog $\rightarrow$ top-K items

Optional Offline Text
user profile P_u
item description/perception D_i

minimal online inputs

Candidate Pool C_u
 $C_u = R_K(H_u)$
or $\{y_u\} + K - 1$ retrieved hard negatives
item text X_C

2 Route

Routing Signals z_u
uncertainty: score entropy / margin
conflict: short-vs-long or score disagreement
noise: sparse, repeated, low-rated history
budget: token / latency constraint

Topology Router r_θ
 $m = r_\theta(H_u, C_u, X_C, z_u, B)$

3 Reason

Linear Protocol
preference $\rightarrow$ match $\rightarrow$ rank

Tree Protocol
split intents/aspects
branch compare $\rightarrow$ aggregate

Graph Protocol
factor nodes + support/conflict edges
message aggregation

select or collapse under budget

4 Rank

Executor
structured reasoning
score each candidate

Output
ranking over C_u
 $\hat{y}_u = \arg \max_{c \in C_u} (c)$

Training / Evaluation Signals

recommendation loss or NDCG reward + router oracle/regret supervision + cost regularization + optional verifier/reward-only item-attribute graph

graph/verifier resources are not assumed as default online prompt inputs